

Ronit Baniya Gupta

Computer Engineer & Builder of Things

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PROFILE

Computer engineer from Pokhara, Nepal. I work with microcontrollers the way other people work with keyboards - and I've been at it since the first time I held an Arduino and was immediately done for. My work lives where code touches real wires: embedded systems, IoT ecosystems, computer vision and the occasional website when the pitch lands. I'd rather ship a rough working thing today than a perfect thing next year. I've taken hardware from breadboard to market - a home-automation switch with a custom PCB, a call-based motor controller that works where WiFi doesn't reach, and an emergency-response ecosystem that's swept the hackathon circuit. If it doesn't exist, I build it.

EDUCATION

- : Bachelor of Computer Engineering - Pokhara University
- : Diploma in Computer Engineering - Batch 2020
- : 3rd DAN Black Belt - Taekwondo
- : High-Performance Coach Trained
- : 3+ Innovation Awards

SKILLS

Languages: Python, C, C++, Java, JavaScript, HTML, CSS, VHDL
Hardware: ESP32, ESP8266, Arduino Uno, Arduino Nano, nRF52840, STM32
Sensors: MPU6050, MAX30102, Pulse Sensor, DS18B20, MLX90614, Ultrasonic, PIR, RFID, IR, GPS, SIM800L
Displays: OLED, LCD I2C, 1.28" Round TFT
Comms: HC-05, HM-10, NRF Modules, BLE, Wi-Fi, MQTT
Power: 18650, TP4056, Buck Converters, LM2596, 7805, Bridge Rectifiers
Software: Firebase, Supabase, React, Node.js, Flutter, OpenCV, MediaPipe, PlatformIO, ESP-IDF, MIT App Inventor

SELECTED PROJECTS

Suraksha360 (2025) - Smart security & emergency response ecosystem
Swasthya Sathi (2025) - Healthcare platform with AI symptom checker
SurakshaGhar (2024) - Home automation I actually shipped
SwiftIgnite (2025) - Keyless motorcycle ignition system
GRAVAS (2025) - Gesture-controlled robotic arm
Air Mouse (2022) - Mouse controlled by pointing

ACHIEVEMENTS

1st - PEC Tech Expo (2022): RUDRA-1 & Suraksha360
1st - GCES IT Expo (2024): Suraksha360
1st - UDAAN 2025 (2025): Suraksha360
2nd Runner-up - OCTATE 2025 (2025): Suraksha360
1st Runner-up - Innosphere 1.0 (2025): Suraksha360 ecosystem
Shipped - SurakshaGhar (2024): SurakshaGhar

PRINCIPLES & EXTRAS

- : Originality over everything - No copy-paste ideas. If someone already made it, I make it better or I don't make it.
- : Build it, then improve it - I'd rather ship a rough working thing today than a perfect thing next year.
- : Brutally honest feedback - Tell me I'm wrong and why. Comforting lies never built anything.
- : 3rd DAN Black Belt in Taekwondo - started BS 2070, stopped, came back, finished high-performance coach training.
- : Weak at theory, strong at shipping. I learn better with a screwdriver in my hand.